Voice over IP (VoIP) & Session Initiation Protocol (SIP)

Redes de Comunicações II

Licenciatura em Engenharia de Computadores e Informática

Prof. Amaro de Sousa (asou@ua.pt)
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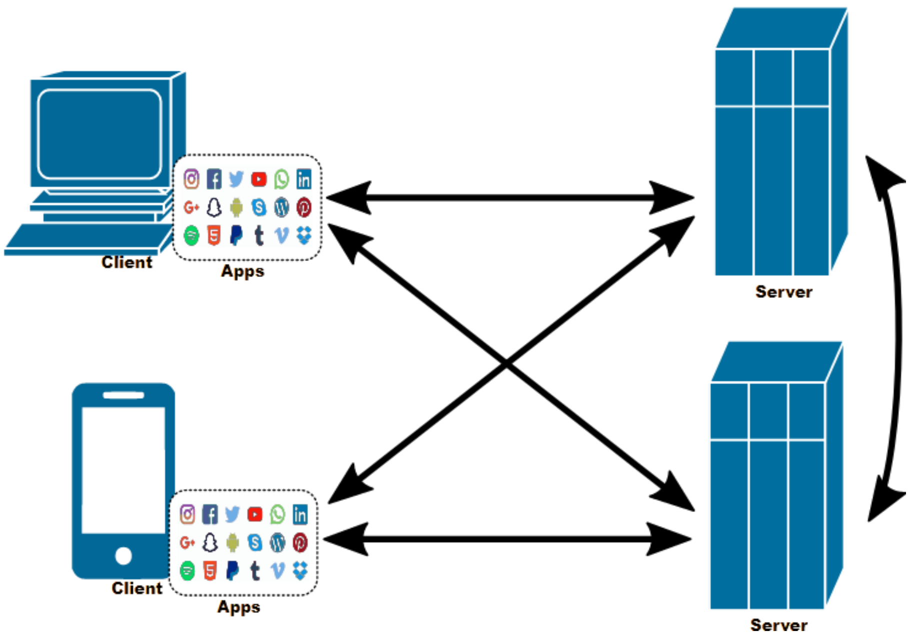
Voice over IP (VoIP)

- Voice over IP (VoIP) is a set of technologies used for voice communication sessions over IP networks.
- How VoIP works:
  1. Analog to Digital: analogue voice is converted in the transmitter (a computer or a VoIP phone) into a digital stream (using a codec) which is broken down into a sequence of data chunks sent in IP packets.
  2. Transmission: IP packets are transmitted over the IP network towards the receiver.
  3. Digital to Analog Reassembly: the receiver reassembles the digital stream converting it back to the analogue voice (with the same codec used in the transmitter).
- Typically, VoIP technologies detect silence periods at the transmitter and do not generate IP packets during these periods (avoiding transmission of "noise")
- VoIP requires session establishment (for example, to agree on the codecs to be used)

Quality of Service (QoS) issues with VoIP

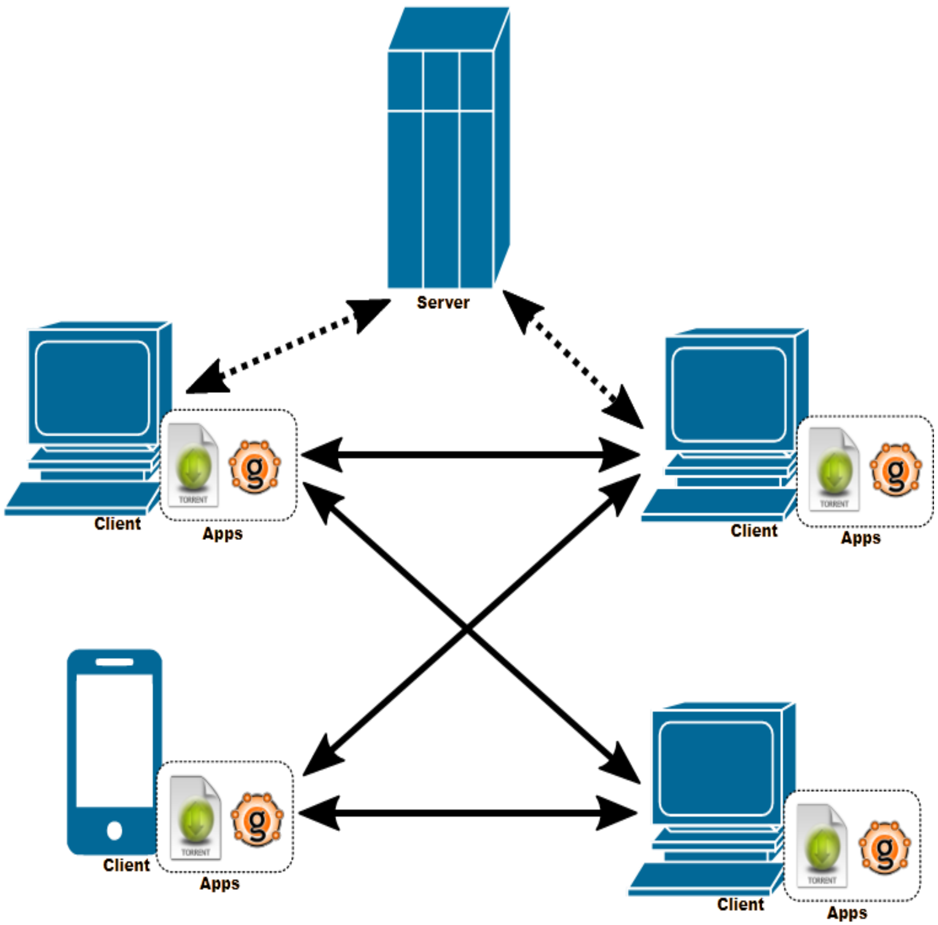
- When transmitted over an IP network, a stream of IP packets of a VoIP session suffer:
  - Packet Delays:
    - due transmission and propagation delays on links,
    - due to switching and queueing delays on routers
  - Packet Loss:
    - due to network congestion (router buffer overflow),
    - due to too late arrival of packets at the receiver for playout.
- Packet Delay tolerance:
  - a one-way delay (mouth-to-ear) should ideally be up to 150 milliseconds for good conversational quality
- Packet Loss tolerance:
  - depending on the used codec, packet loss rates between 0.1% and 1% can be tolerated

Services running a Client-Server Model



- Servers:
  - Always ON (i.e. always available to accept communications from clients)
  - IP address is always known (usually through a DNS name)
  - May communicate between them (i.e., may act as clients between them).
- Clients:
  - Communicate with servers (i.e., they initiate the communications with servers).
  - Not always ON (i.e., clients are ON only when in operation).
  - May have dynamic addresses.
  - They do not communicate between themselves.

Services running a Peer-to-Peer (P2P) Model



- Clients:
  - Communicate between themselves.
  - Can be ON only when in operation.
  - May have dynamic addresses.
  - Peer discovery may be done within the P2P network or using central servers.
- Servers:
  - Usually, they exist to support the operation of the P2P service.

Session Initiation Protocol (SIP)

- SIP is defined by RFC 3261, June 2002.
- SIP was designed for creating, modifying and terminating sessions between two or more participants.
  - Not limited to VoIP calls (can be multimedia, video-conference).
- SIP can be transported over UDP or TCP protocols.
- It is an alternative to the complex H.323 family of protocols.
  - Due to its simplicity, SIP became much more popular than H.323.
- It provides a peer-to-peer service between endpoints. Peers in a session are called user agents (UAs):
  - User-agent client (UAC): a client application that initiates the request of a SIP session.
  - User-agent server (UAS): a server application that receives and reacts to SIP session requests.
  - A SIP endpoint works both as a UAC and a UAS.

SIP functionalities

- SIP supports five facets of establishing and terminating multimedia communications:
  - User location - determination of the end system to be used for communication
  - User availability - determination of the willingness of the called party to engage in communications
  - User capabilities - determination of the media (audio/video) and media parameters to be used
  - Session setup - "ringing", establishment of session parameters at both called and calling party
  - Session management - including transfer and termination of sessions, modifying session parameters, and invoking services

SIP as a component of a multimedia architecture

- SIP does not provide an integrated communications system.
- SIP is a component to be used with other protocols (to set up a complete multimedia architecture), such as:
  - Real-time Transport Protocol (RTP) for transporting real-time data (audio/video) over IP networks and providing QoS feedback
  - Real-Time Streaming Protocol (RTSP) for controlling the delivery of streaming media over IP networks
  - Media Gateway Control Protocol (MEGACO) for controlling gateways to the Public Switched Telephone Network (PSTN)
  - Session Description Protocol (SDP) for describing multimedia sessions.
- SIP is to be used in combination with other protocols to provide complete services to users, but the basic functionalities and operation of SIP do not depend on any of the other protocols.

SIP clients and SIP servers

- SIP Clients
  - Phones (software based, or hardware) and Gateways
  - User Agents, that can act as a
    - Client when they send a SIP request (UAC),
    - Server when they react to a received SIP request (UAS).
- SIP Servers
  - Proxy server
    - Receives SIP requests from a client and forwards them on the client's behalf.
    - Receives SIP messages and forward them to the next SIP server in the network.
    - Provides functions such as authentication, authorization, network access control, routing, reliable request retransmission, and security.
  - Redirect server
    - Provides the client with information about the next hop that a SIP message should take (then, the client contacts the next-hop server or UAS directly).
  - Registrar server
    - Register current location of each active UAC.
    - Registrar servers are often co-located with a redirect or proxy server.

SIP messages

- SIP is used for Peer-to-Peer Communications (voice/video data is exchanged directly between SIP terminals), though it was proposed to work as a Client-Server Model (to set up the communications between terminals).
- SIP is a text-based protocol (similar to HTTP) and uses the UTF-8 charset. A SIP message is either a **Request** from a client to a server, or a **Response** from a server to a client.
  - A Request message consists of a **Request-Line**, one or more Header Fields, an empty line (indicating the end of the Header Fields), and an optional message-body;
  - A Response message consists of a **Status-Line**, one or more Header Fields, an empty line (indicating the end of the Header Fields), and an optional message-body.
  - Similarly to HTTP, all lines (including empty ones) are terminated by a "carriage-return" "line-feed" character sequence (CRLF).

SIP Request messages

- SIP uses SIP Uniform Resource Indicator (URI) to indicate the requested user or service. The general form of a Request-URI is:
  - `sip:user:password@host:port;uri-parameters`
    - `sip:John@doe.com`
    - `sip:+14085551212:pass1234@195.2.43.29:5060`
    - `sip:alice@atlanta.com;maddr=239.255.255.1;ttl=15`
- A Request message contains one of a possible set of methods:
  - RFC 3261 defines six methods: INVITE, ACK, OPTIONS, BYE, CANCEL, and REGISTER.
  - SIP extensions provide additional methods: SUBSCRIBE, NOTIFY, PUBLISH, MESSAGE, ...
- The Request-Line of a Request message includes a Method, a Request-URI, and SIP-Version separated by a single space:
  - Example: `INVITE sip:John@doe.com SIP/2.0`

Methods in SIP Request messages

Basic Methods:

- INVITE – to establish a session
- ACK – to confirm an INVITE request
- BYE – to end a session
- CANCEL – to cancel establishing a session
- REGISTER – to indicate the current terminal location
- OPTIONS – to communicate information about the capabilities of the calling and callee terminals

Some Extended Methods:

- SUBSCRIBE – to subscribe a notification service in the server
- NOTIFY – to notify the subscriber terminal of a new event
- PUBLISH – to publish an event into the server
- MESSAGE – to transport Instant Messages

SIP Response messages

SIP Request messages are replied with SIP Response messages, whose Status-Line is a code of 3 digits. There are six classes of codes:

- 1xx – Informational responses, examples:
  - `100 Trying`
  - `180 Ringing`
- 2xx – Success responses, example:
  - `200 OK`
- 3xx – Redirection responses
- 4xx – Request failures, examples:
  - `401 Unauthorized`
  - `403 Forbidden`
- 5xx – Server errors
- 6xx – Global failures

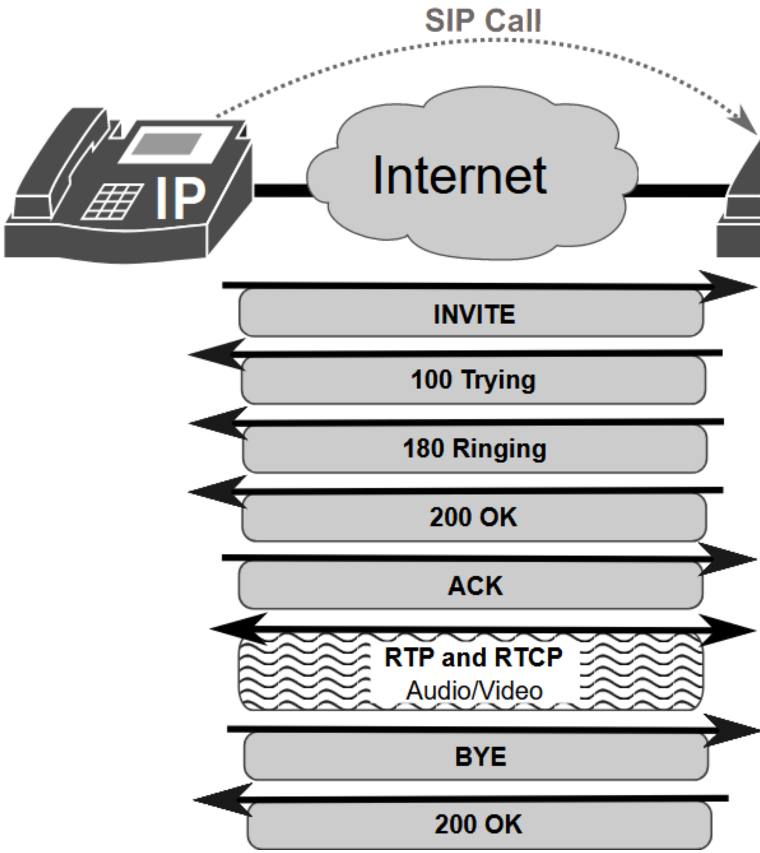
Session Description Protocol (SDP)

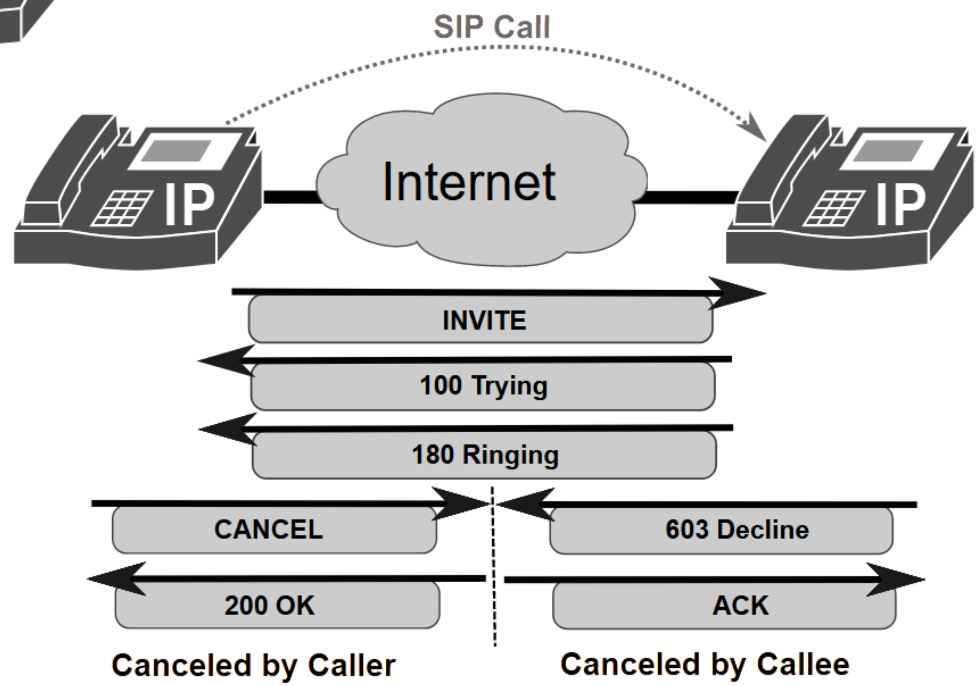
- SDP is defined by RFC 4566 (July 2006) and its core purpose is parameter negotiation concerning media communications.
- For two devices to establish a direct VoIP call or video conference over the internet, they need to resolve different compatibility issues:
  - Where should the data go? (IP addresses and Port numbers)
  - How should the data be coded? (which codecs like G.711, or H.264 are supported)
  - What type of media is being sent? (audio, video, or data streams)
- SDP is a standardized text-based format used to describe these capabilities.
- SDP allows devices to agree on network locations, codecs, and formats before real-time media streams start being sent via transport protocols like RTP or RSTP.

Session Description Protocol (SDP)

- SDP works dynamically using a peer-to-peer negotiation framework known as the Offer-Answer Model:
  - **The Offer:** The caller generates an SDP text block enumerating all the media types, formats, and codecs their device can support, along with the network ports they have opened to receive incoming data.
  - **The Answer:** The callee reads the offer, compares it to its own capabilities, selects the highest mutually supported codec, and returns its own SDP answer containing its respective network addresses and ports.
- Once this exchange completes, both terminals have a valid matching configuration to open direct peer-to-peer data channels.
- SDP is typically carried inside the messages of signalling protocols, such as SIP for IP telephony or WebRTC APIs for browser-based video calls.

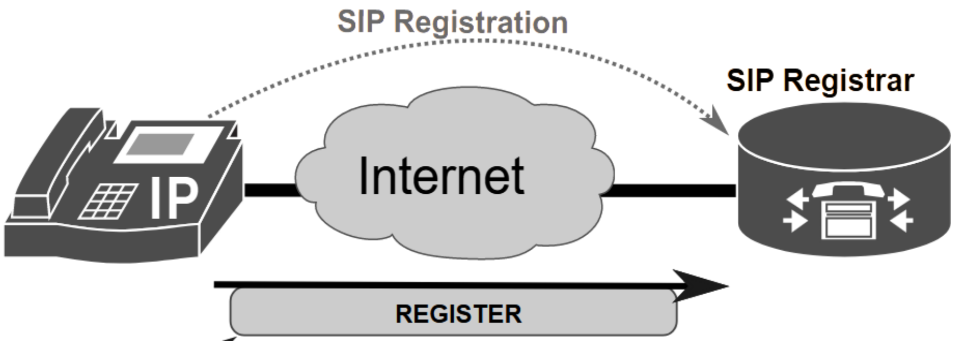
SIP signalling in direct calls





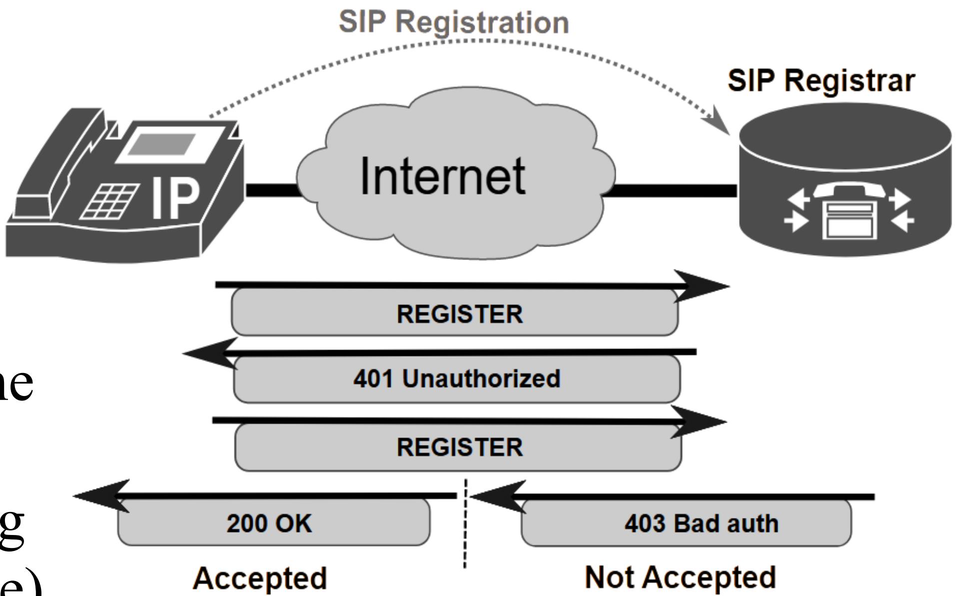
- SDP of the caller is sent in the message body of the INVITE message
- SDP of the callee is sent in the message body of the 200 OK message
- Issues:
  - The caller must know the IP address, port number and transport protocol of the callee
  - It fails if the callee has a private IP address (i.e., behind a NAT gateway)

SIP Registrar server



- SIP Registrar servers store the location of SIP endpoints.
- A user with a valid AOR (Address of Record) sends a REGISTER message with its AOR and its current location. Example (simplified):
  - `REGISTER sip:John@doe.com SIP/2.0`
  - `From: <sip:John@doe.com>`
  - `Contact: <sip:John@192.168.56.100:5060;transport=udp>`
  - `Expires: 3600`
- The header field "`Expires:`" of the REGISTER message defines for how much time (in seconds) the registration should be valid.
  - A REGISTER message with "`Expires: 0`" deregisters the endpoint.
- SIP Proxy servers query SIP Registrar servers for routing information (i.e., when they need to forwards SIP messages to the SIP endpoint).

SIP Registrar server

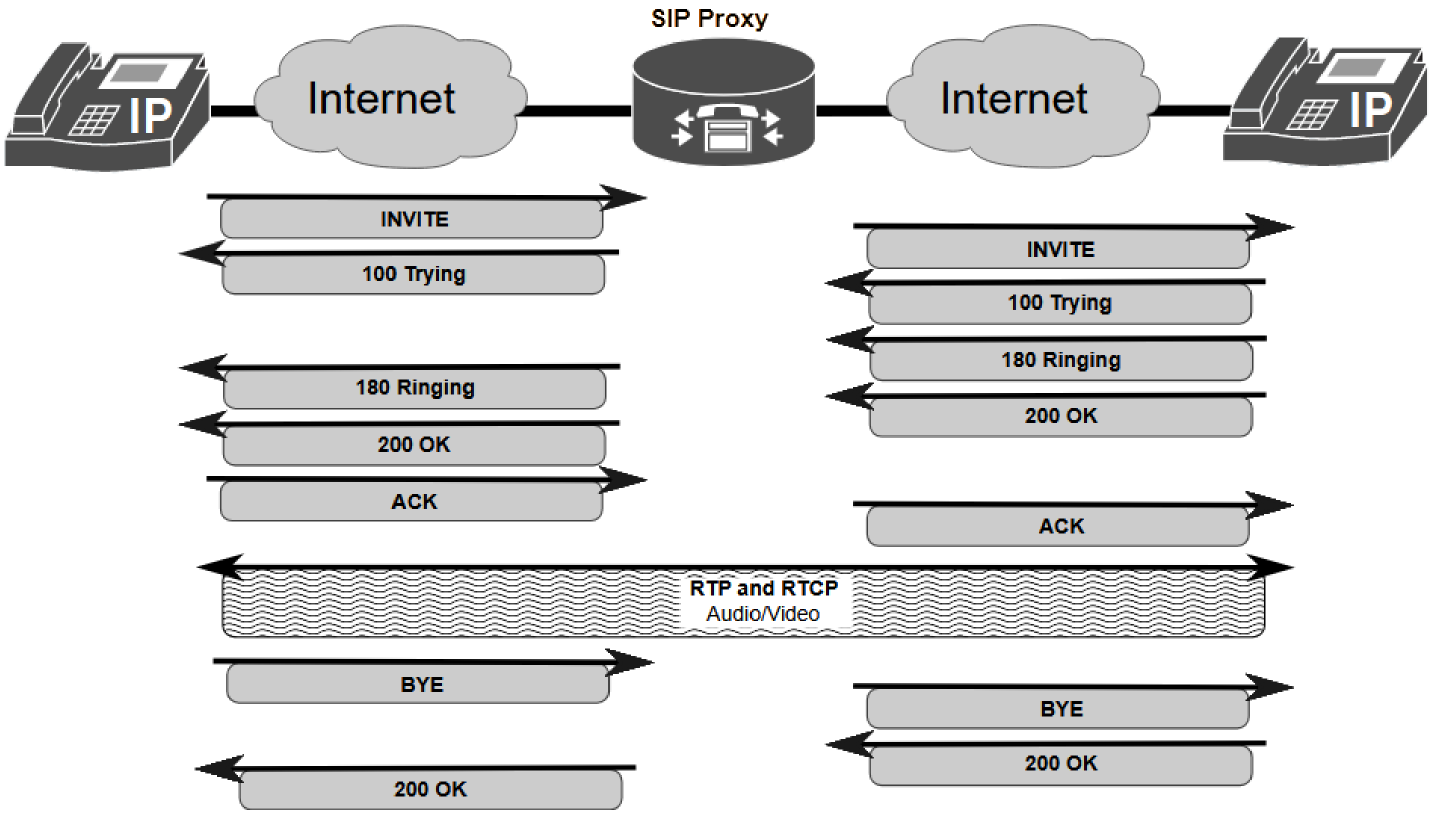


- Registration usually requires authentication.
- The first REGISTER message does not include authentication.
- The SIP Registrar responds to the first REGISTER message with "`401 Unauthorized`" providing a cryptographic challenge (nonce).
- The SIP endpoint hashes its password with the challenge data and sends a second REGISTER message with the hash provided in the header field "`Authorization:`".
- The SIP Registrar either accepts the registration with the status line "`200 OK`" or rejects the registration with the status line "`403 Forbidden (Bad auth)`".

SIP Registrar server: some registration rejection reasons

- **Wrong Password**
- **Location Blocking:** VoIP providers can block traffic from some foreign countries to prevent fraud. If you travel or use a VPN, the server denies you.
- **IP Access Control Lists (ACL):** In enterprise networks, the VoIP service may be configured to only allow registrations from specific office subnets. An employee trying to connect from home without a VPN cannot register.
- **Suspended/Deactivated Account:** The SIP credentials match, but the account has been disabled due to an unpaid bill or a suspected security breach.
- **Max Registration Limit Reached:** If the system allows only one device per user, and a desk phone is already registered, a softphone (running in a mobile phone) trying to register the same user will be rejected.

SIP signalling using Proxy server



- In this case, the users of both SIP endpoints have previously registered in the same domain:
  - For example: from `John.Smith@doe.com` to `Ed.Andersson@doe.com`

  and, so, the Proxy server of the domain can obtain from the Registrar server the IP address and port number of the callee SIP endpoint.

Locating SIP servers

- When the callee of a SIP session (identified by a SIP URI) is not in the same domain of the caller domain, the SIP server of the caller must identify the SIP server of the callee:
  - the caller's SIP server needs to know the Transport protocol, IP address and Port number to be used to forward SIP messages to the callee's SIP server.
- This information is obtained by the DNS service using the **Name Authority Pointer (NAPTR)** and the **Service (SRV)** DNS records:
  - NAPTR DNS record provides a mapping from a domain name to an SRV record and a specific transport protocol.
  - SRV DNS record provides the name(s) of the responsible domain server(s).

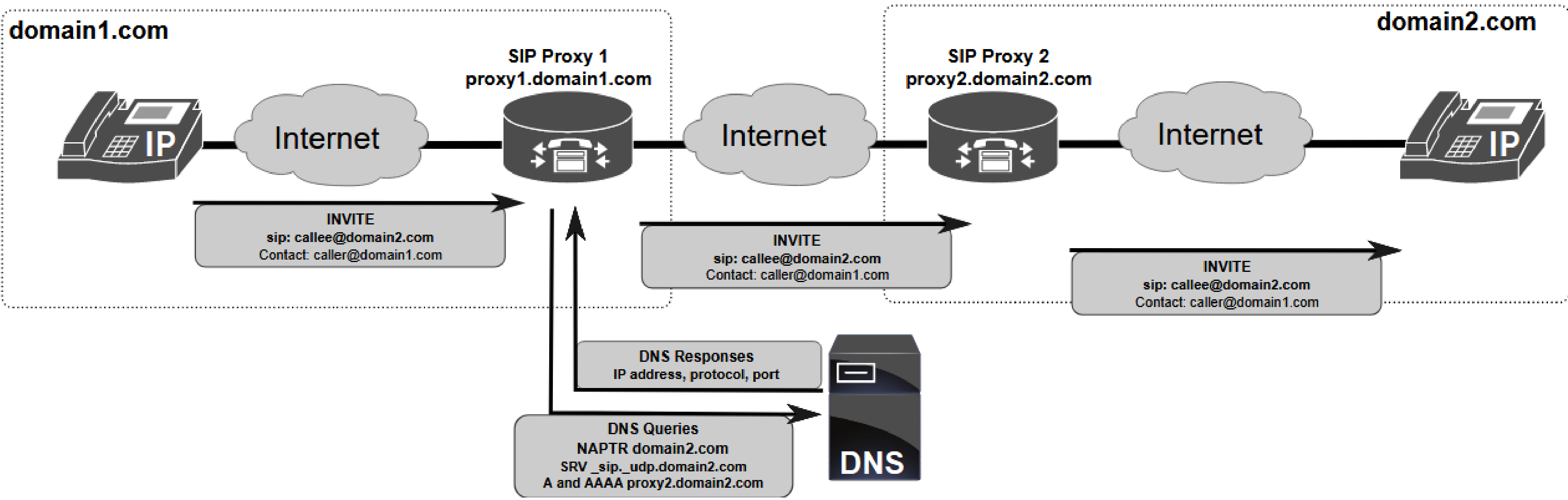
Locating SIP servers (example)

- A server wishes to resolve `sip:user@example.com`
- First, it sends a NAPTR DNS query for domain "`example.com`" and the DNS reply is:
  - `IN NAPTR 100  50  "s"  "SIP+D2U"  ""  _sip._udp.example.com`

  it obtains UDP as the transport protocol to be used.
- Then, it sends an SRV DNS query for "`_sip._udp.example.com`" and the DNS reply is:
  - `IN SRV  0  1  5060  server1.example.com`
  - `IN SRV  0  2  5060  server2.example.com`

  it obtains two possible server names and the Port number used by each of them (in this case, 5060 for both).
- Finally, it sends A and/or AAAA queries for the chosen server name to obtain its IPv4 and/or IPv6 addresses.

SIP Proxy forwarding illustration

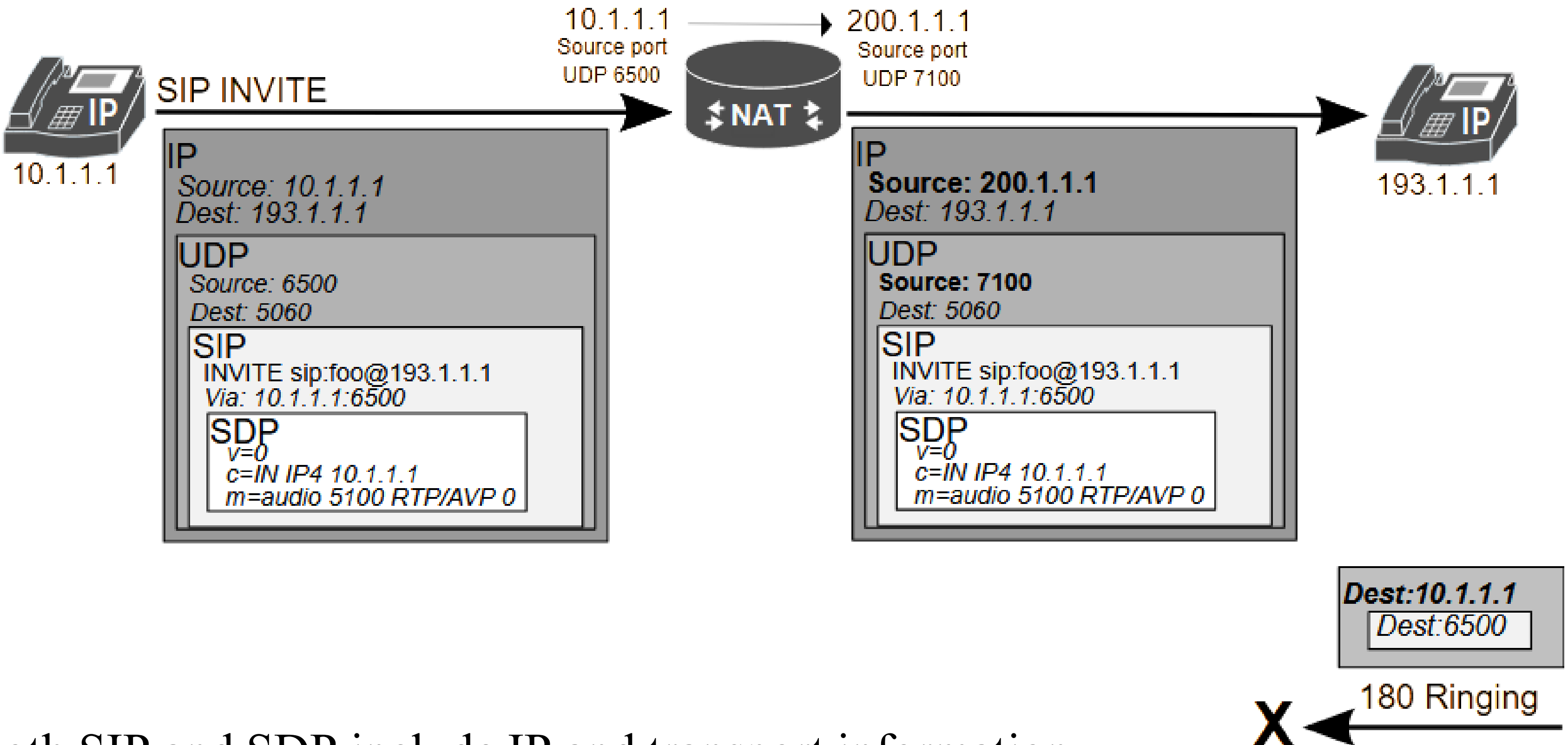


- In this case, the user of each SIP endpoint has previously registered on its domain.
- When the INVITE reaches SIP Proxy 2, it obtains from its Registrar server the IP address of its SIP endpoint.

SIP and NAPT

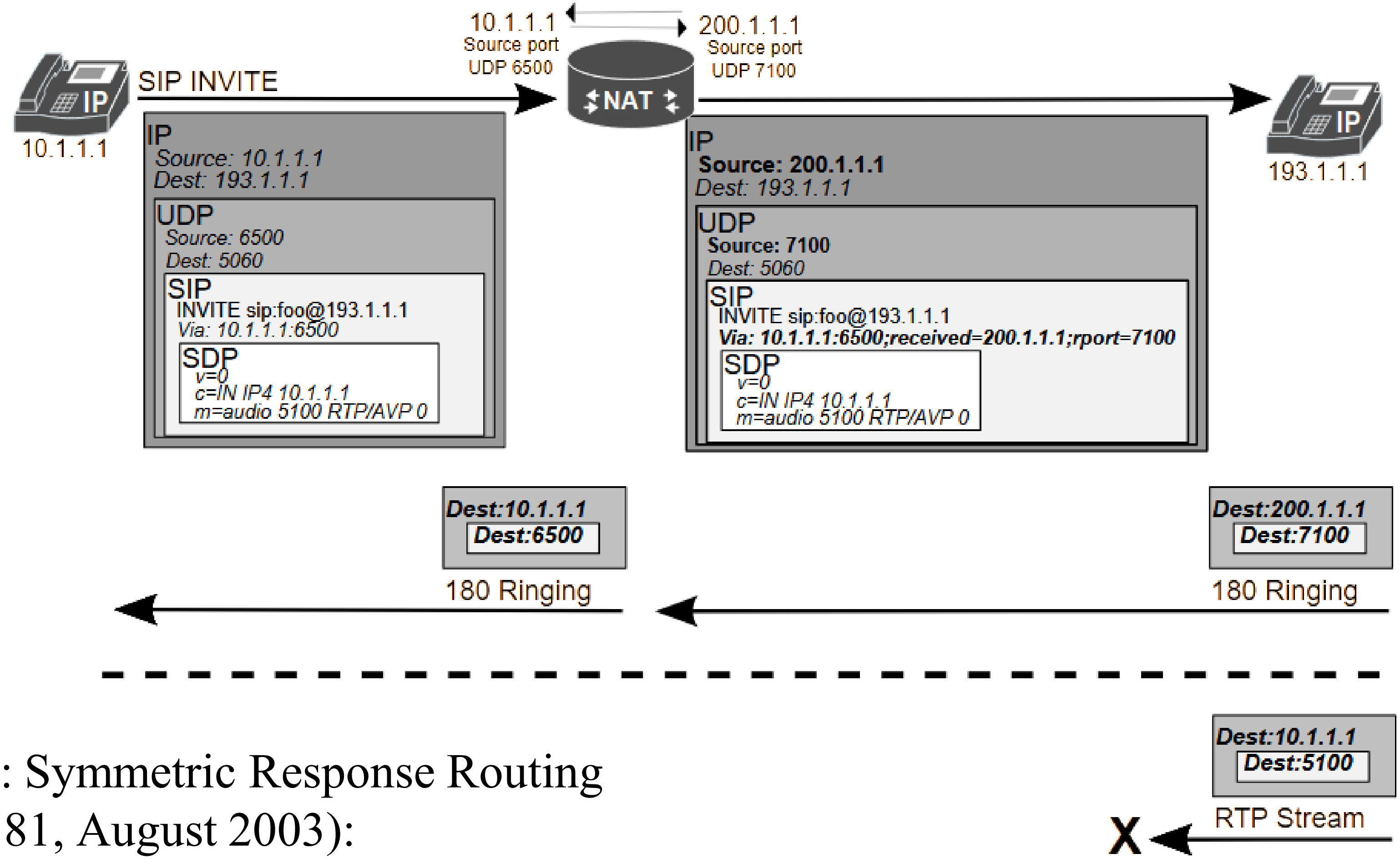
- SIP works fine when all SIP hosts are in a public domain IP network.
- When SIP clients are in a NAT domain, there is an architectural conflict between SIP and NAPT (Network Address Port Translation) protocol:
  - SIP is an application layer protocol which embeds IP addresses and port numbers inside the application payload (the message body)
  - By default, NATP only translates IP addresses and port numbers in the IP and transport (UPD or TCP) headers from the private domain to the public domain.
- Because standard NAT routers do not change data in the application layer, the public domain SIP servers receive (from the SIP clients) requests with private IP addresses instead of public IP addresses.

SIP and NAPT



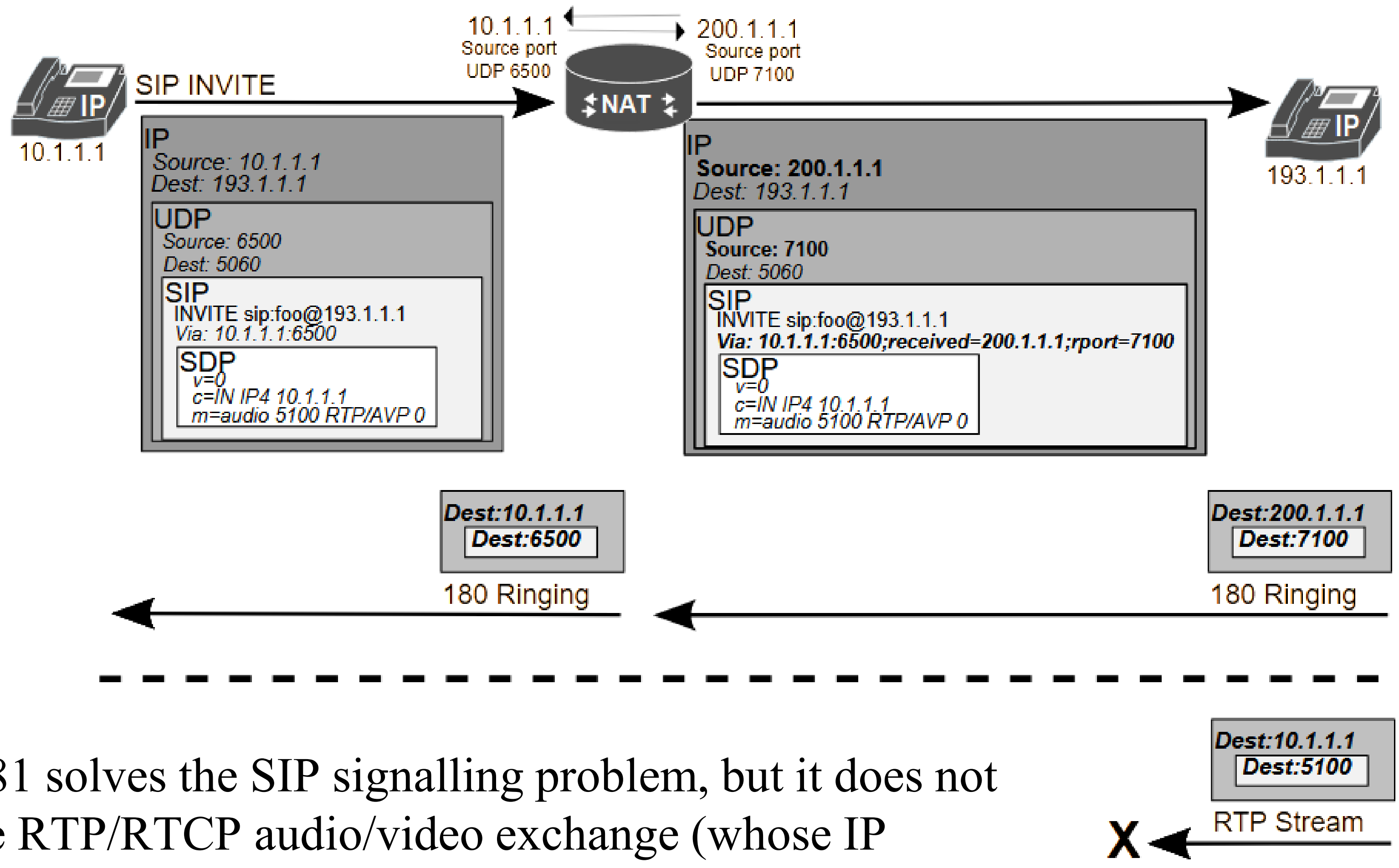
- Both SIP and SDP include IP and transport information
- By default, if the caller is in a private IP network, the NAT translation does not allow successful SIP signalling because:
  - the caller information in the Header Field "`Via:`" of the SIP Request message is used to reply with the SIP Response message of the callee.

SIP and NAPT: solution for the SIP signalling



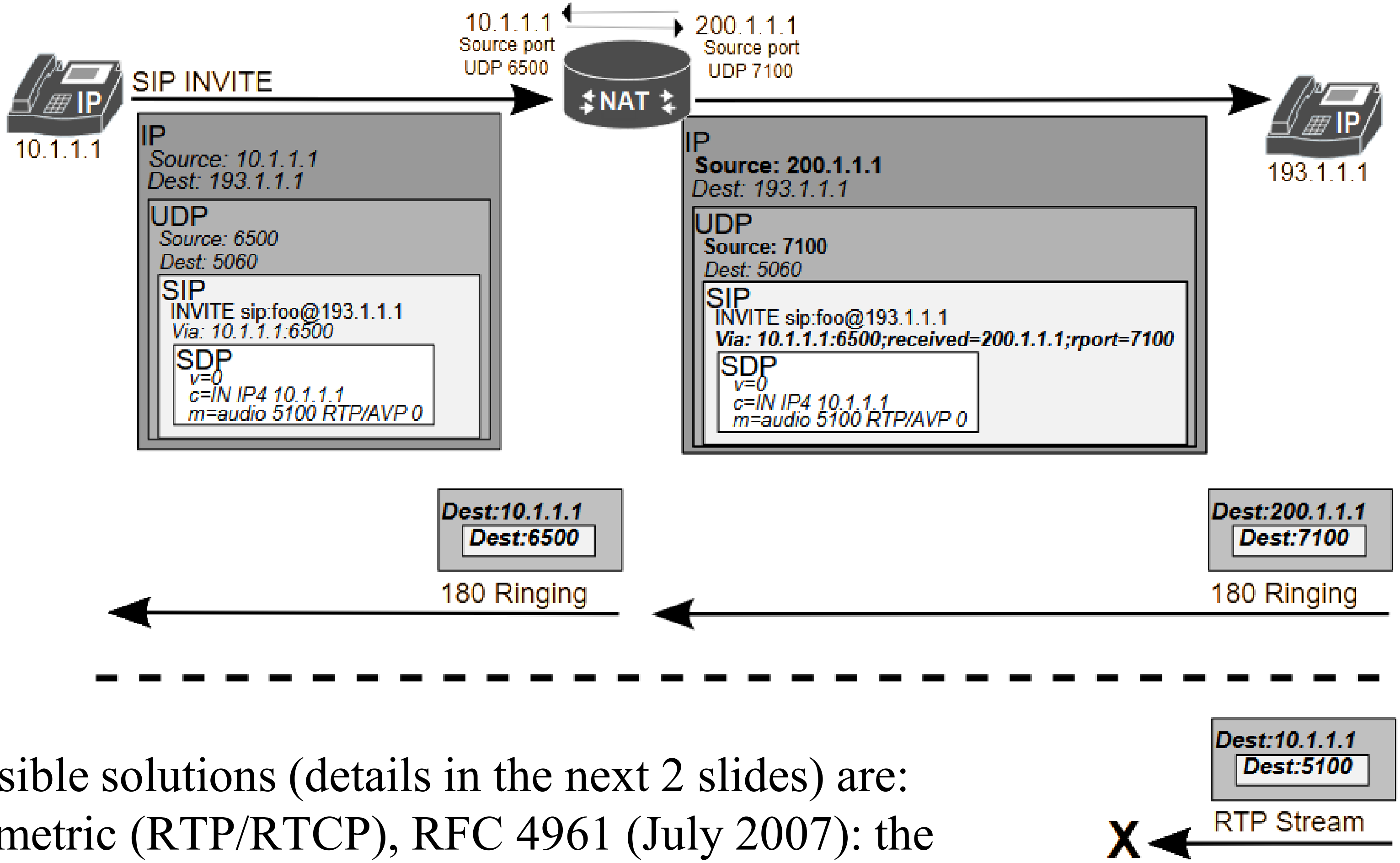
- Solution: Symmetric Response Routing (RFC 3581, August 2003):
  - SIP payload is also "translated" by NAT gateway, adding a **received** and **rport** fields with the public IP address/port number.

SIP and NAPT: solution for the SIP signalling



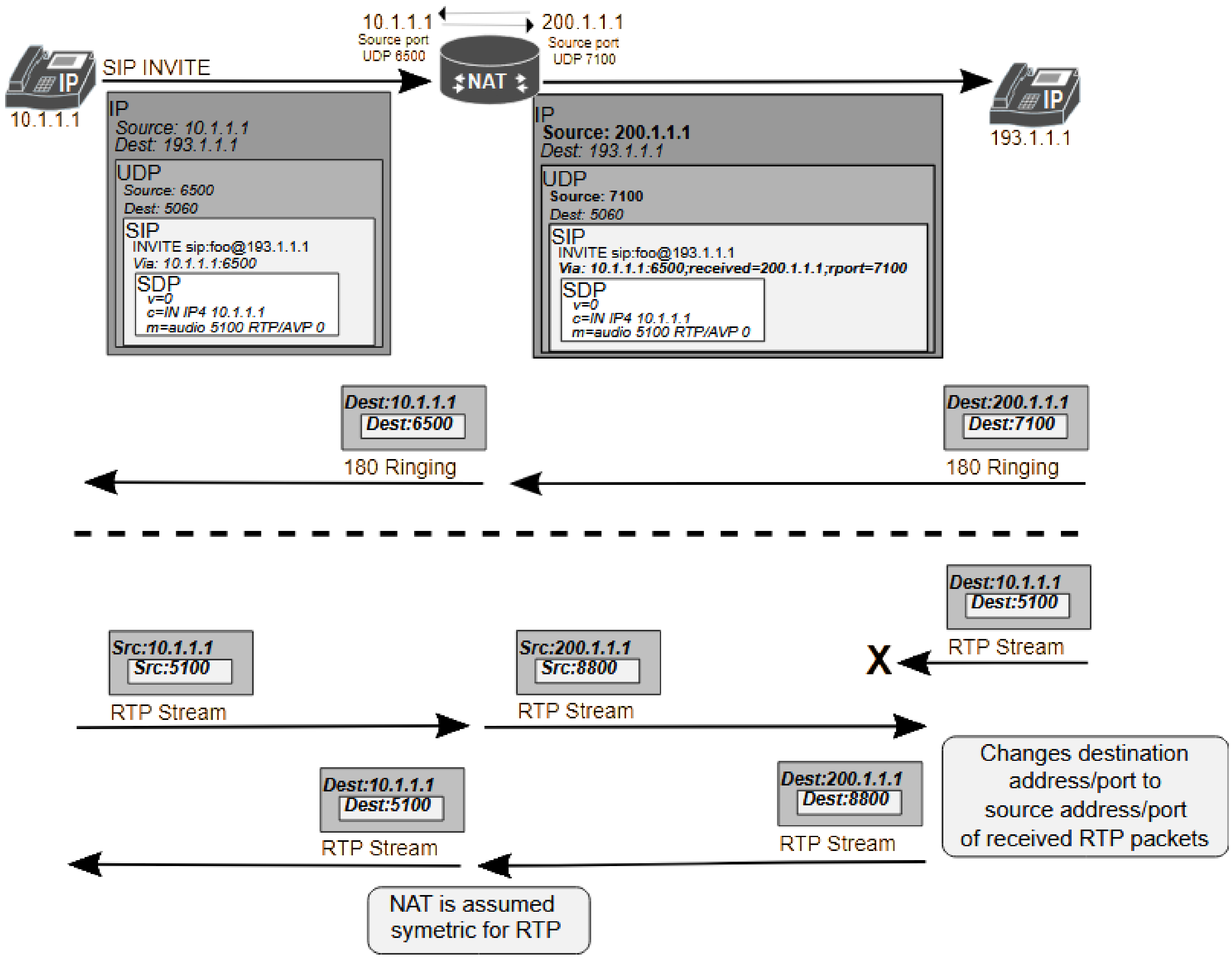
- RFC 3581 solves the SIP signalling problem, but it does not solve the RTP/RTCP audio/video exchange (whose IP addresses to be used are described in SDP).

SIP and NAPT: solution for the SIP signalling

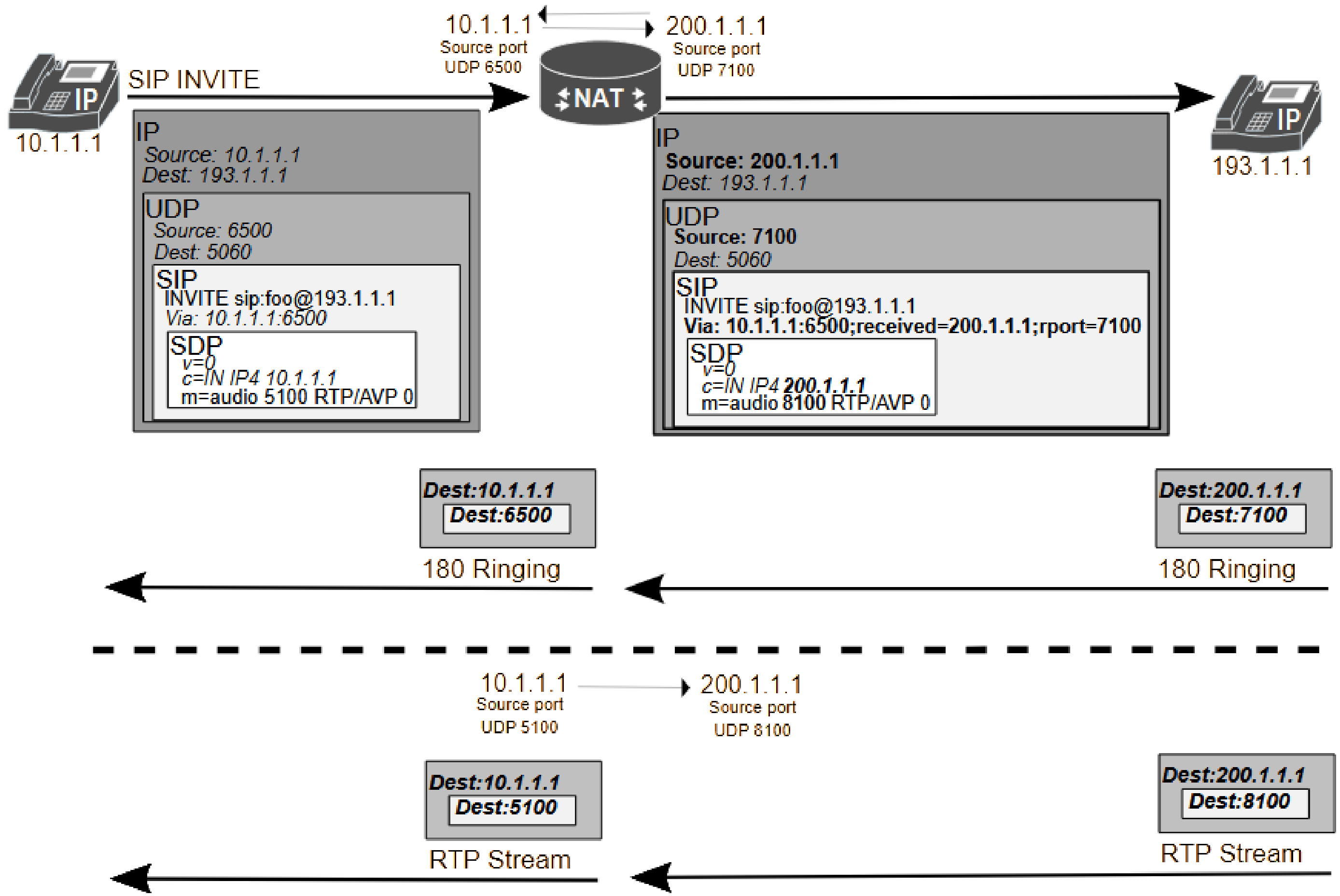


- Two possible solutions (details in the next 2 slides) are:
  - Symmetric (RTP/RTCP), RFC 4961 (July 2007): the source port number used in RTP/RTCP is equal to the destination port number announced by the receiver (through SDP)
  - NAT SIP Application Layer Gateway (ALG): translation is also done in RTP/RTCP packets

Symmetric (RTP/RTCP) NAT



NAT SIP Application Layer Gateway (ALG)



- Requires NAT gateway to translate all RTP/RTCP data packets.
- It imposes a heavy computational burden on NAT gateway.